

Stock Buy and Sell

Difficulty: Medium | Accuracy: 29.18% | Submissions: 311K+ | Points: 4 | Average Time: 20m

Given an array `arr[]` denoting the cost of stock on each day, the task is to find the maximum total profit if we can buy and sell the stocks any number of times.

Note: We can only sell a stock which we have bought earlier and only one instance of stock can be traded.

Examples

Input: `arr[] = [100, 180, 260, 310, 40, 535, 695]`

Output: 865

Explanation: Buy the stock on day 0 and sell it on day 3 $\Rightarrow 310 - 100 = 210$. Buy the stock on day 4 and sell it on day 6 $\Rightarrow 695 - 40 = 655$. Maximum Profit = $210 + 655 = 865$

Input: `arr[] = [4, 2, 2, 2, 4]`

Output: 2

Explanation: Buy the stock on day 3 and sell it on day 4 $\Rightarrow 4 - 2 = 2$

Input: `arr[] = [4, 2]`

Output: 0

Explanation: Don't Buy the stock.

Constraints

$2 \leq \text{arr.size()} \leq 10^6$

$0 \leq \text{arr}[i] \leq 10^6$

Topic Tags

Arrays, Dynamic Programming, Greedy

Company Tags

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